CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

PART A (80 MARKS)

QUESTION 1

a) Explain two formal security models used for access control, stating the specific security concern each model is designed to address.

(2 marks)

b) State six points about the access control matrix, access control lists and capability lists as access control structures, covering what each structure represents and one advantage or disadvantage of each.

(6 marks)

QUESTION 2

a) Explain how NIDS sensors should be positioned in a network, giving the typical sensor locations.

(2 marks)

b) Describe six preventive network security mechanisms that an organization can deploy to protect its network, explaining the role each mechanism plays.

(6 marks)

QUESTION 3

a) State TWO (2) facts about the chain of trust in the Trusted Computing architecture, including how measurements are recorded and where the chain begins.

(2 marks)

b) Explain the roles of the two primary system components of the Next Generation Secure Computing Base (NGSCB), describing what each component is and how it supports trusted operation of the platform.

(6 marks)

QUESTION 4

a) State TWO (2) differences between morals and laws as described in the lecture on ethics and computer security.

(2 marks)

b) State SIX (6) professional conduct requirements that the ACM code expects a computing professional to follow when carrying out computing work.

(6 marks)

QUESTION 5

a) Explain two characteristics of Trojan horse malware that distinguish its behaviour on an infected host.

(2 marks)

a) Discuss six malware attack techniques and propagation methods, explaining how each technique operates and the impact it has on an infected system or its users.

(6 marks)

QUESTION 6

a) State the four goals of access control and identify the component that makes authorization decisions.

(2 marks)

b) Refer to your answer in (a). Explain six points about the formal security models used for access control, covering the Bell-La Padula model, the Biba model and the Chinese Wall model, including the security concern each addresses and the rules that govern access under the Chinese Wall model.

(6 marks)

QUESTION 7

a) List TWO (2) key points about software insecurity and buffer overflows as presented in the Software Security lecture.

(2 marks)

b) Compare the security mechanisms found in Windows with those found in UNIX, focusing on the strengths and weaknesses of each. Provide six (6) distinct points covering user account and object management, access control, administration requirements, and missing features.

(6 marks)

QUESTION 8

a) List two classes of threats and, for each, state which security objective (confidentiality, integrity or availability) is affected and which service counters it.

(2 marks)

b) The Samy worm spread through MySpace by exploiting a cross-site scripting flaw. Explain six aspects of this incident, covering how the JavaScript was injected, how the worm propagated through user profiles and friend lists, and the scale of the resulting infection.

(6 marks)

QUESTION 9

a) Explain the two stages of the biometric authentication process and state what happens during each stage.

(2 marks)

b) A bank plans to deploy biometric authentication at its branch counters. Explain the biometric mechanisms and technologies involved, covering how biometrics are defined, the categories of biometric applications, and the characteristics and drawbacks of specific technologies.

(6 marks)

QUESTION 10

a) Explain two password security controls that a system administrator can apply to protect password files and accounts.

(2 marks)

b) Describe six aspects of single sign-on and Kerberos as authentication mechanisms in large distributed environments, covering the motivation for single sign-on, the solutions available, the design criteria of Kerberos, and how a user logs in to Kerberos.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) An organisation deploys an intrusion detection system to defend its network against malware and other attacks. Explain five aspects of how such detection and defence mechanisms work, covering what the IDS does, how misuse detection recognises attacks, and the limitations of these approaches.

(5 marks)

b) State five distinct threat classes or attack techniques that can compromise a computer system, and for each one identify a countermeasure or defence mechanism that addresses it.

(5 marks)

QUESTION 2

a) Explain five ethical obligations that the ACM code expects a computing professional to uphold when developing and deploying technology, and explain why such a professional code may not always be enforceable.

(5 marks)

b) Discuss the ethical concerns raised by the staff member's action and the company's responsibility, referring to privacy, the ACM code of ethics, and the nature of professional ethical codes.

(5 marks)

END OF QUESTION PAPER